Wigner Resolution from Action Capacity

Brian S. Mulloy
Independent Researcher, Corktown, Detroit, Michigan, USA[*]
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Phase-space formulations of quantum mechanics usually ignore that resolution is bounded below by Planck's quantum of action. In the Wigner formulation, the justification is that even though pointwise resolution forces the quasi-probability negative, those values carry physical structure. Zurek located that structure at scales set by the state's action capacity. We identify those scales as the principal axes of a family of de Gosson quantum blobs. Convolving the Wigner function to the scale of a single $h/2$ blob restores resolution to Planck's quantum of action and renders it non-negative. The state's elliptical Heisenberg cell of action capacity $\pi\,\Delta x\,\Delta p$ circumscribes the family, each blob deforming at constant action $h/2$. Integrating over the family reaches the resolution $\pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the Heisenberg cell. The resulting portrait is everywhere non-negative and preserves the structure of the Wigner function in the same locations. This resolution carries no free parameter and sharpens as the action capacity grows.

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The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise, below Planck's quantum of action, and Hudson's theorem forces it negative there [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6], while a non-negative portrait admits a probability interpretation [7–10] and supports classical sampling [11]. Action capacity $\pi\,\Delta x\,\Delta p$ deforms a family of $h/2$ quantum blobs that resolve the Wigner function to $\pi\hbar^2/(\Delta x\,\Delta p)$, the resolution symplectic polar duality sets on any non-negative portrait of the state.

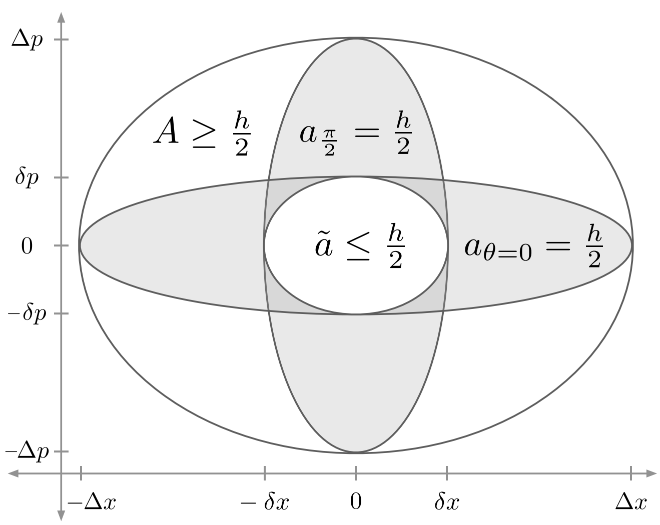

FIG. 1. **Heisenberg cell, principal quantum blobs, and quorum cell.** The Heisenberg cell is the state's covariance ellipsoid with action capacity $A = \pi\,\Delta x\,\Delta p \geq h/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The quorum cell $\tilde{a}$ is an ellipse with semi-axes $\delta x, \delta p$ and action $\tilde{a} = \pi\,\delta x\,\delta p$. A circumscribes $a_{\theta=0}$ and $a_{\pi/2}$, which circumscribe $\tilde{a}$.

Planck divided phase space into elementary regions of

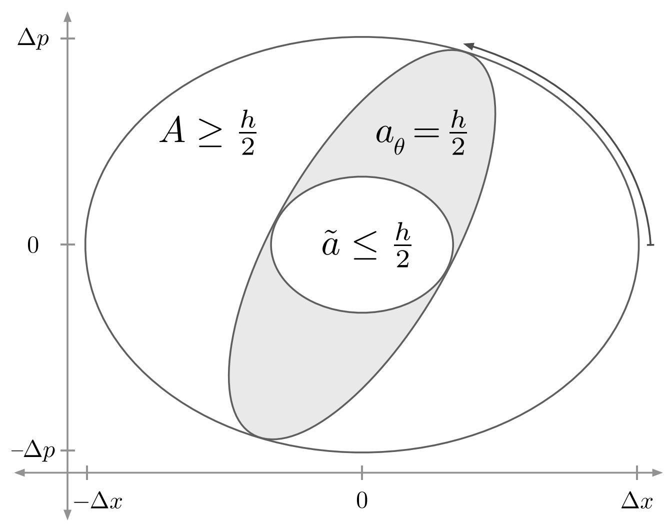

FIG. 2. **Quadrature family of quantum blobs and quorum cell.** The Heisenberg cell A circumscribes the quantum blob family $\{a_\theta\}$, which circumscribes the quorum cell $\tilde{a}$. Across θ the family deforms at constant action $a_\theta = h/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $a_{\theta=0}$ and $a_{\pi/2}$ from Fig. 1.

action [12]. Heisenberg's uncertainty principle $\sigma_x\,\sigma_p \geq \hbar/2$ [13–15] fixes the limit. Geometrically, de Gosson gave this action region the form of the state's covariance ellipsoid, with symplectic capacity $\pi\,\Delta x\,\Delta p$ [16] and semi-axes $\Delta x = \sqrt{2}\,\sigma_x$ and $\Delta p = \sqrt{2}\,\sigma_p$. This is the Heisenberg cell of action capacity (Figs. 1, 2),

$$A = \pi\,\Delta x\,\Delta p \geq h/2. \tag{1}$$

The Wigner function carries structure at small scales. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate those scales from the state's action capacity [17]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$a_{\theta=0} = \pi\,\Delta x\,\delta p = h/2, \qquad a_{\pi/2} = \pi\,\delta x\,\Delta p = h/2. \tag{2}$$

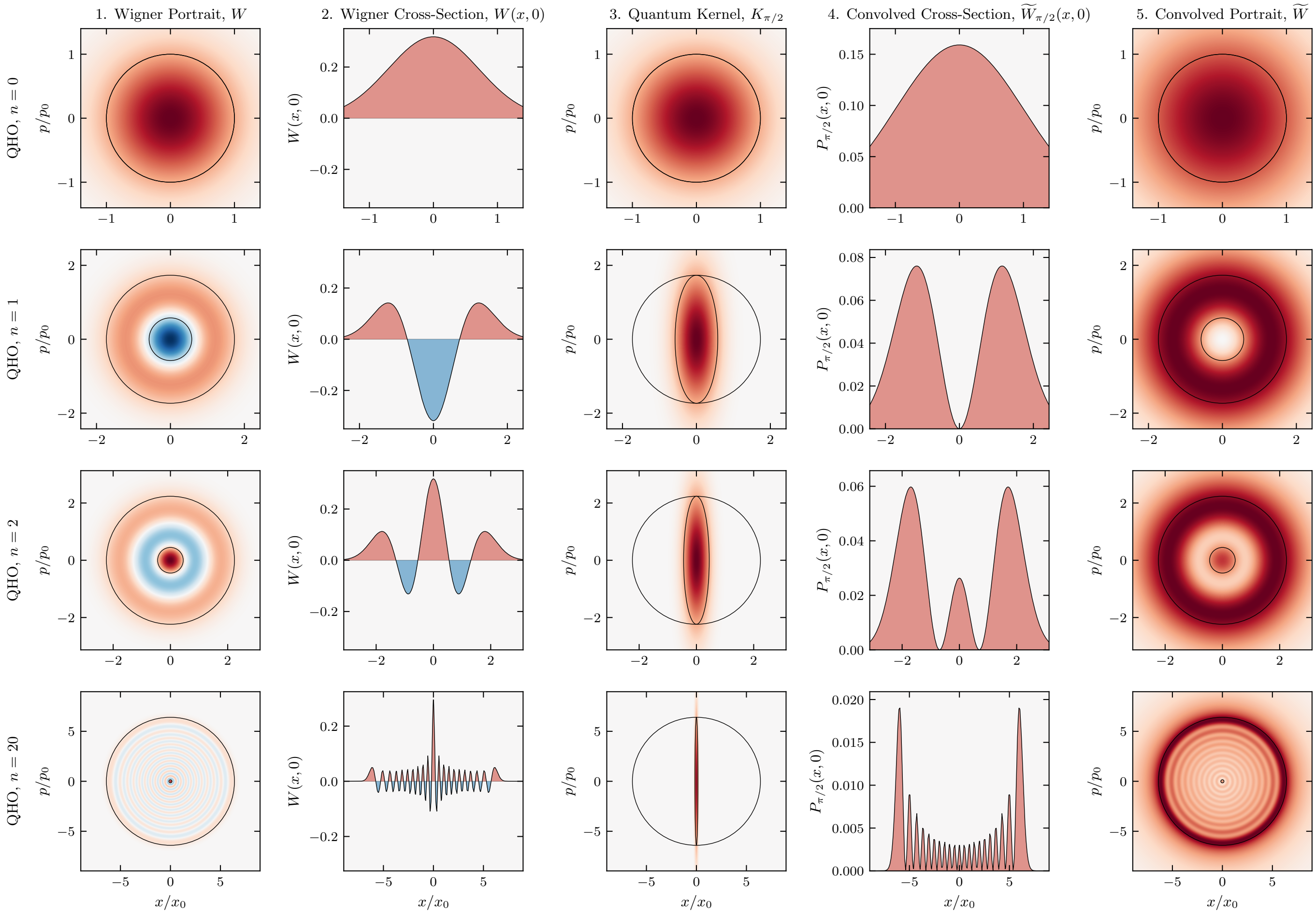

FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(h/2) = 2n + 1$. Column 1: Wigner function $W(x, p)$ (negativity in blue) overlaid with the Heisenberg cell A and quorum cell $\tilde{a}$. Centered at the covariance origin, $\tilde{a}$ outlines the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x, 0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $a_{\pi/2}$ (action $h/2$, restoring resolution up to Planck's quantum of action). Column 4: Convolved cross-section $\widetilde{W}_{\pi/2}(x, 0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\widetilde{W}(x, p)$, integrated over the blob family across all angles, overlaid with A and $\tilde{a}$ (symplectic polar duals). It is everywhere non-negative, with positive lobes matching the count and position of those in W, resolved at scale $\tilde{a}$. For the vacuum ($n = 0$), A and $\tilde{a}$ coincide as a Gaussian blob at $A = h/2$. As n increases, A grows and $\tilde{a}$ reciprocally shrinks ($\tilde{a}\,A = (h/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving $\tilde{a}$ as a fine central cell. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

The capacity $\Delta x, \Delta p$ runs along the major axes, the resolution $\delta x, \delta p$ along the minor.

Between these principal members the quantum blob a_θ deforms across $\theta \in [0, \pi)$ at constant action $h/2$, the preimage of the rigid ellipse $\hat{a}_\theta$ under the scaling $X = x/\Delta x$, $P = p/\Delta p$,

$$a_\theta = \{(x, p) : (X, P) \in \hat{a}_\theta\}. \tag{3}$$

The rotated quadratures $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$ are the homodyne quadratures of optical tomography at phase θ. In the scaled coordinates $\hat{a}_\theta$ is the rotation by θ of a single ellipse,

$$\hat{a}_\theta = \left\{(X, P) : X_\theta^2 + \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2 \le 1\right\}. \tag{4}$$

The Heisenberg cell A circumscribes each blob a_θ.

The Wigner function of the state is

$$W(x, p) = \frac{1}{\pi\hbar}\int \psi^*(x + s)\,\psi(x - s)\,e^{2ips/\hbar}\,ds. \tag{5}$$

The Wigner function of each quantum blob a_θ is the centered Gaussian whose $h/2$ ellipse is a_θ,

$$K_\theta(x, p) = \frac{1}{\pi\hbar}\exp\!\Big[-X_\theta^2 - \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2\Big]. \tag{6}$$

The convolution of W with each quantum kernel K_θ is

$$\widetilde{W}_\theta(x, p) = (W * K_\theta)(x, p). \tag{7}$$

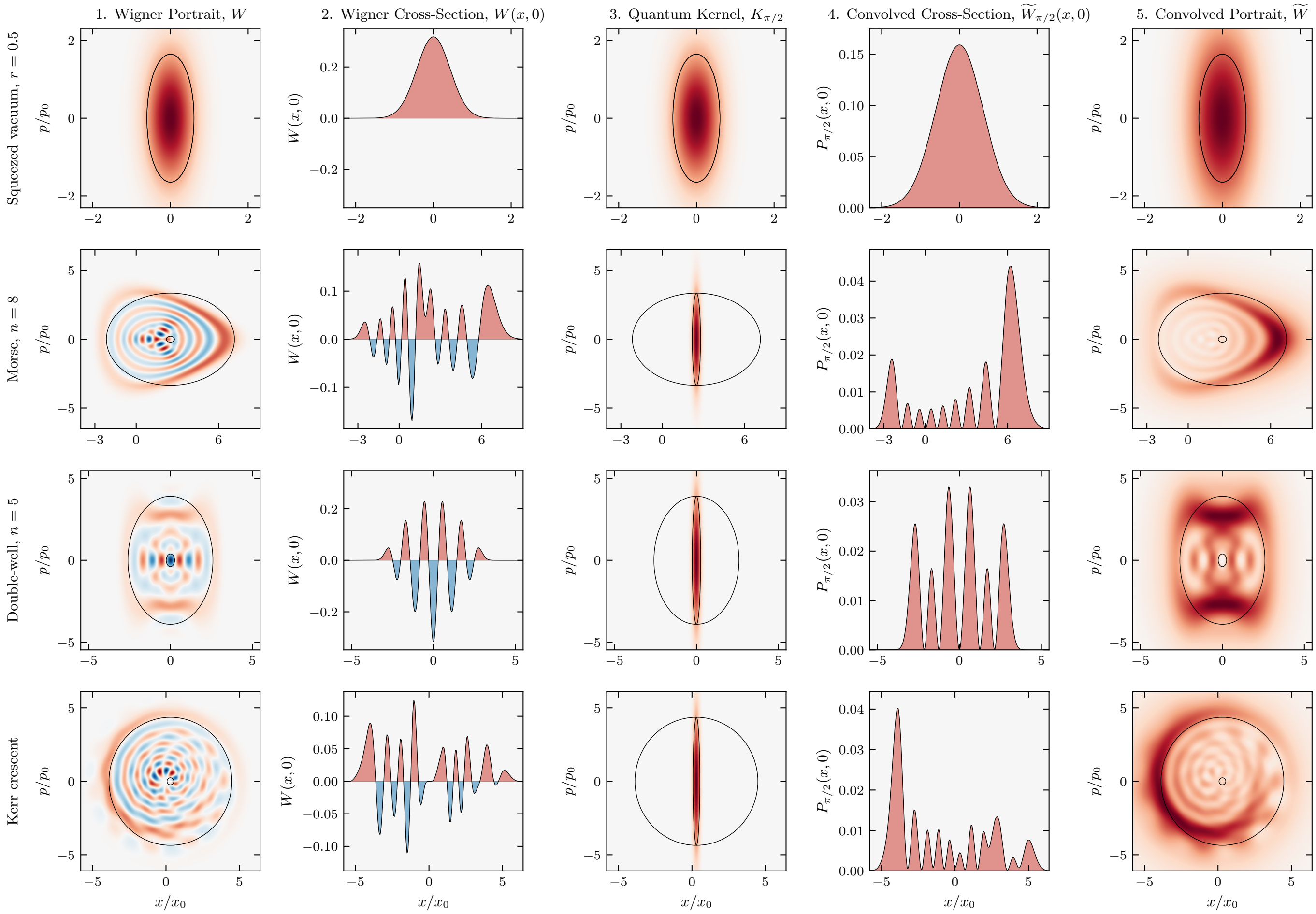

FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the cell overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at the scale of $\tilde{a}$.

Convolving with a Gaussian of action $h/2$ lifts W up to Planck's quantum of action and yields a non-negative function [8, 10, 18], so $\widetilde{W}_\theta$ is non-negative everywhere.

Zurek also located the scale $a \sim \hbar \times (\hbar/A)$ [17] of the chessboard structure in the compass state (Fig. 5, column 1). We identify its rotationally invariant form as the quorum cell $\tilde{a}$. Named after the quorum of observables in optical tomography [19], $\tilde{a}$ is the ellipse with semi-axes $\delta x, \delta p$ (Fig. 1) and action

$$\tilde{a} = \pi\, \delta x\, \delta p = \frac{\pi \hbar^2}{\Delta x\, \Delta p}. \tag{8}$$

$\tilde{a}$ is the scale of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1). The quorum cell $\tilde{a}$ is the symplectic polar dual of the Heisenberg cell A [20, 21]. The duality is reflexive, so A and $\tilde{a}$ each fix the other, and antimonotone, so $\tilde{a}$ shrinks as A grows. Each blob a_θ is circumscribed by A and circumscribes $\tilde{a}$, nesting as $A \supseteq a_\theta \supseteq \tilde{a}$ (Fig. 2). A quorum of these a_θ blobs, each respecting $h/2$, reaches $\tilde{a}$ together. Resolution and capacity are reciprocal at the quantum of action, $\tilde{a}\, A = (h/2)^2$.

Integrating $\widetilde{W}_\theta$ over the family gives the phase-space portrait of the state at the resolution $\tilde{a}$,

$$\widetilde{W}(x,p) = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta(x,p)\, d\theta. \tag{9}$$

$\widetilde{W}_\theta \geq 0$ for every θ, so $\widetilde{W} \geq 0$ everywhere. Tomographic reconstruction inverts marginals pointwise to recover W [22–24]. Over the same θ, convolving W with the quantum kernel family $\{K_\theta\}$ builds the non-negative portrait $\widetilde{W}$.

A quantum blob a_θ too large for A to circumscribe, or too small to circumscribe $\tilde{a}$, is a blob of another state and portrays that state, not this one [20, 25]. Because every blob a_θ of this state circumscribes the quorum cell $\tilde{a}$, and a non-negative portrait $\widetilde{W}$ smooths W by a Gaussian

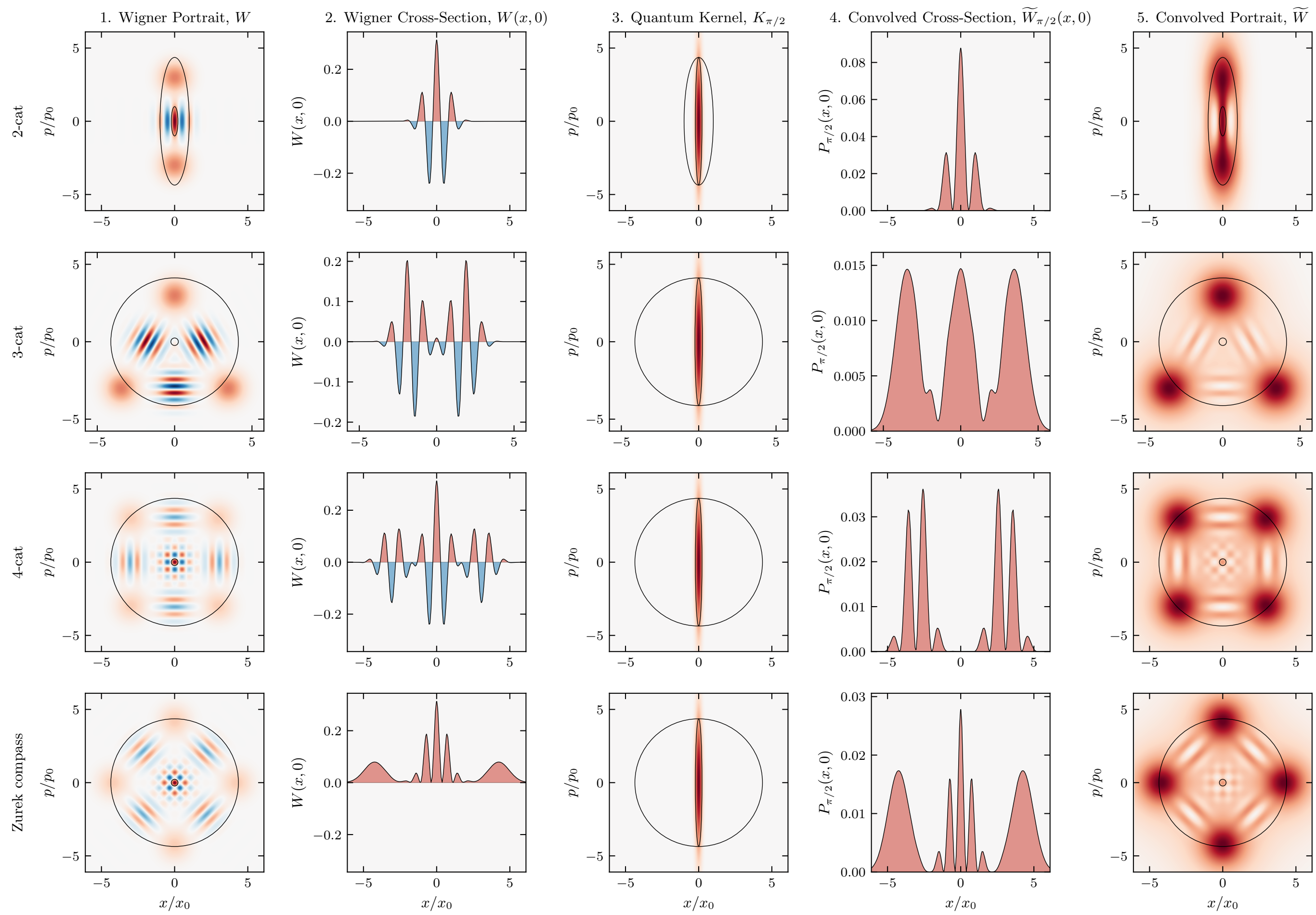

FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\widetilde{W}$ (Column 5) is everywhere non-negative, with positive lobes matching the count and location of those in W, resolved at scale $\tilde{a}$. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

of action at least $h/2$ [8, 10], $\tilde{a}$ is the finest resolution any non-negative portrait attains, and the family $\{a_\theta\}$ reaches it across $\theta \in [0, \pi)$.

The principal frame, where the Heisenberg cell A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [18], so the map that diagonalizes a state's covariance carries it to this frame at preserved action. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely. In higher dimensions, action $h/2$ no longer singles out a blob [25], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

A fixed vacuum-width Gaussian [7] matches the state only at the vacuum $A \sim h/2$ and erases structure at resolution $\tilde{a}$ for larger states. Methods with a free parameter [26–28] can resolve more finely, the scale chosen as a parameter. The family here carries no free parameter, fixed by the action capacity A alone.

Figures 3, 4, and 5 show numerical results across twelve states [29]. The portraits W and $\widetilde{W}$ share the same structure at scale $\tilde{a}$, with $\widetilde{W}$ verified non-negative.

The uncertainty product $\sigma_x\sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [30] and Bohr [31] identified as requiring special care. The quantum blob does not. As A grows, the action of each a_θ is fixed at $h/2$. Squeezing [32] reaches $\sigma_x\sigma_p = \hbar/2$ by external apparatus. Here it is inherent to the state. A Schrödinger cat with $\Delta x \sim 10$ cm and $\Delta p \sim 1$ kg·m/s has positional resolution $\delta x = \hbar/\Delta p \sim 10^{-34}$ m, vastly smaller than the radioactive atom that triggers its apparatus [33]. The scale called sub-Planck is the minor axis δ of an $h/2$ blob. The correspondence principle follows from $A \to \infty$, the physical limit underlying the heuristic $h \to 0$.

The negative Wigner function W and its non-negative portrait $\widetilde{W}$ have their structure at a single scale $\tilde{a}$, set by the state's action capacity A. Zurek scales and de Gosson quantum blobs give the resolution. Heisenberg's product gives the capacity. As the action capacity grows to $\pi\,\Delta x\,\Delta p \geq h/2$ the resolution sharpens to $\pi\hbar^2/(\Delta x\,\Delta p) \leq h/2$. We read this structure as the geometry of phase space itself, which de Gosson takes as prior to the uncertainty principle [34].

This research received no external funding. Numerical computations were performed with QuTiP [35] and SciPy [36]. The code reproducing the numerical results, together with Supplemental Material mapping this Letter onto Zurek and de Gosson, is openly available [29].

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* bmulloy@umich.edu

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